# Experiment Report: `spec27_d512_4b_k2` (130K)

Extended Proxy Tape Training — L1 Eta Collapse Under Prolonged Proxy Gradients

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**Abstract**

This report documents the `spec27_d512_4b_k2` extended experiment: a 130K-step run (30K initial + 100K warm restart) using Spec 27's per-level tape strategy (L0=exact, L1=proxy) on the standard d=512, 4-block, k=2 CMS configuration. The run extends the previously validated 30K proxy experiment to test long-horizon proxy training.

The headline finding is **L1 eta collapse**: after step ∼57K, Block 0's L1 output gate drops from ∼0.95 to oscillating 0.15–0.75 (mean ∼0.45–0.55), while L0 eta saturates to 1.0. The model is progressively suppressing L1's output contribution under prolonged proxy training. Despite this, loss continues to improve (smoothed 3.94 at 30K → 2.60 at 129K), meaning the model adapts by routing information through L0.

Key results:

- **Loss**: 11.0 → 2.72 (phase 1, 30K), warm restart → 2.60 (phase 2, 129K).
- **L1 eta collapse**: Block 0 L1 eta drops from 0.96 to ∼0.48 after step 57K.
- **L1 alpha floor-pinning**: L1 alpha hits the 0.80 floor in Blocks 0/1/2 by step 60K.
- **Block 2 L1 theta**: Continues rising from 0.45 (30K) to 0.88 (125K) — the sole block where proxy L1 actively learns.
- **Block CV collapse**: Depth specialization drops from 0.55 to 0.18 — blocks homogenize.
- **Throughput**: 1,186 tok/s mean, consistent across both phases.
- **18 gradient spikes**, including a 36.6-norm event at step 117,584.

## Contents

**1 Experiment Configuration** **3**
1.1 Two-Phase Structure . . . 3

**2 Loss Trajectory** **4**
2.1 Comparison with spec25 (First 30K) . . . 5

**3 Gradient Analysis** **5**
3.1 Depth Specialization Collapse . . . 7
3.2 Gradient Spikes . . . 8

**4 Gate Diagnostics: The Headline Finding** **8**
4.1 L1 Eta Collapse (The Headline) . . . 8
4.2 L1 Alpha Floor-Pinning . . . 9
4.3 L0 Eta Saturation . . . 9
4.4 Full Gate Evolution . . . 10
4.5 Block 2 L1 Theta: The Persistent Anomaly . . . 11

**5 Detailed Gate Comparison Table** **11**

**6 Throughput** **12**

**7 Conclusions** **12**

7.1 Extended Proxy Training Produces Adaptive L1 Suppression . . . . . 12
7.2 Block 2: The Exception That Proves the Rule . . . . . 12
7.3 Implications for k=4 Production . . . . . 12
7.4 Depth Specialization at d=512 . . . . . 13
7.5 Next Steps . . . . . 13

# 1 Experiment Configuration

Table 1: Run configuration. Warm restart at step 30K with fresh 100K cosine schedule.

| **Parameter** | **Value** |
|---|---|
| *Architecture* | |
| d_model | 512 |
| num_heads | 8 |
| n_blocks | 4 |
| params | 55,141,386 |
| k / chunk_sizes | 2 / [1, 8] |
| composition | MAG |
| memory_rule | Titans LMM |
| *Tape Strategy (Spec 27)* | |
| tape_strategies | [exact, proxy] |
| tape_multiplier | 1 |
| tape_device | GPU |
| *Training — Phase 1 (steps 0–30K)* | |
| steps | 30,000 |
| lr / warmup | 3e-4 / 500 |
| optimizer | AdamW (GPU stacked) |
| alpha_floor | [0.8, 0.8] |
| seed | 42 |
| data | Dolmino-100B |
| *Training — Phase 2 (warm restart, steps 30K–130K)* | |
| steps | 100,000 |
| lr / warmup | 3e-4 / 500 (fresh cosine schedule) |
| checkpoint | Phase 1 final (step 30K) |

## 1.1 Two-Phase Structure

The run consists of two build phases logged in the metrics:

1. **Phase 1** (0–30K): Initial training from scratch. Cosine schedule over 30K steps. Final loss: 2.715, throughput: 1,182 tok/s, wall time: 3.61h.

2. **Phase 2** (30K–130K): Warm restart from the phase 1 checkpoint with a fresh 100K-step cosine schedule. The LR resets to 3e-4, causing a temporary loss spike at step 30K (smoothed loss jumps from $\sim$3.0 to $\sim$3.9) before the model re-adapts.

The warm restart is visible in every metric as a discontinuity at step 30K: loss spikes, gradient norms spike, and gate values shift. All analysis in this report accounts for this phase boundary.

## 2 Loss Trajectory

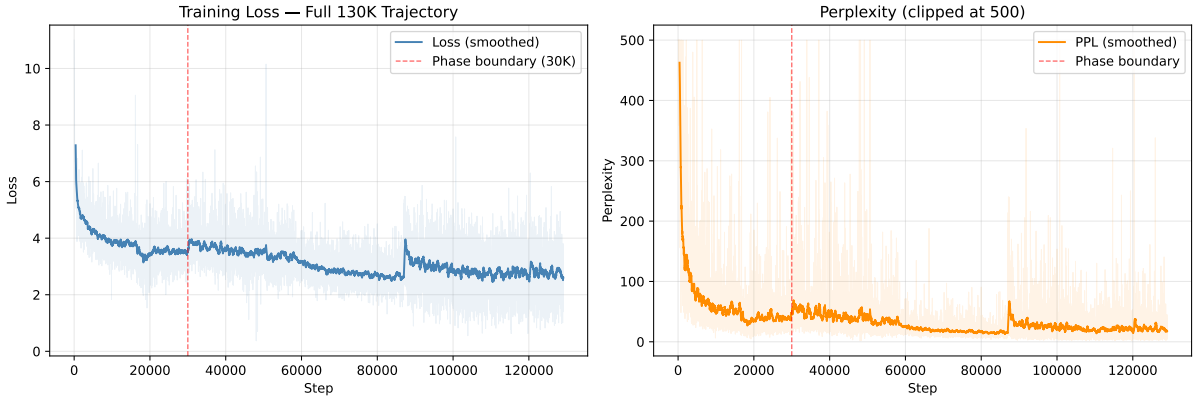


Figure 1: Full 130K loss and perplexity. The phase boundary at 30K is marked by a red dashed line. Loss spikes at 30K due to the warm restart LR reset, then resumes descent. Final smoothed loss: 2.60.

Table 2: Loss progression at key milestones (last-50 smoothed averages).

| Step | Smoothed Loss | Phase |
|---:|---:|---|
| 1,000 | 4.88 | Phase 1 (early) |
| 5,000 | 4.23 | Phase 1 (warmup complete) |
| 10,000 | 3.91 | Phase 1 (mid) |
| 20,000 | 3.41 | Phase 1 (late) |
| 30,000 | 3.94 | Phase boundary (LR spike) |
| 50,000 | 3.63 | Phase 2 (re-adapting) |
| 75,000 | 2.73 | Phase 2 (converging) |
| 100,000 | 2.82 | Phase 2 (plateau/oscillation) |
| 125,000 | 2.77 | Phase 2 (late) |
| 129,000 | 2.60 | Phase 2 (final) |

Notable features:

- The loss at step 100K (2.82) is *worse* than step 75K (2.73), suggesting the cosine schedule's declining LR creates a temporary plateau before the final descent.
- Phase 2 achieves a lower final loss (2.60) than phase 1 (2.72) despite the warm restart disruption.
- The model recovers from the LR spike in ∼20K steps (by step 50K), then improves further.

## 2.1 Comparison with spec25 (First 30K)

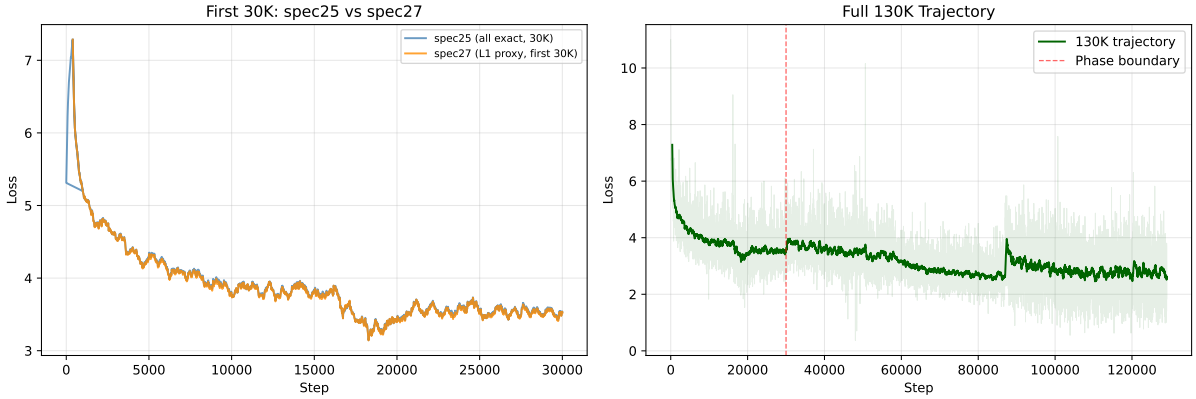


Figure 2: Left: First 30K overlay comparing spec25 (all exact) with this run's first phase. Right: the full 130K trajectory.

The first 30K of this run replicates the spec25 result: loss curves are virtually indistinguishable, confirming the 30K proxy validation holds even when the run continues beyond.

# 3 Gradient Analysis

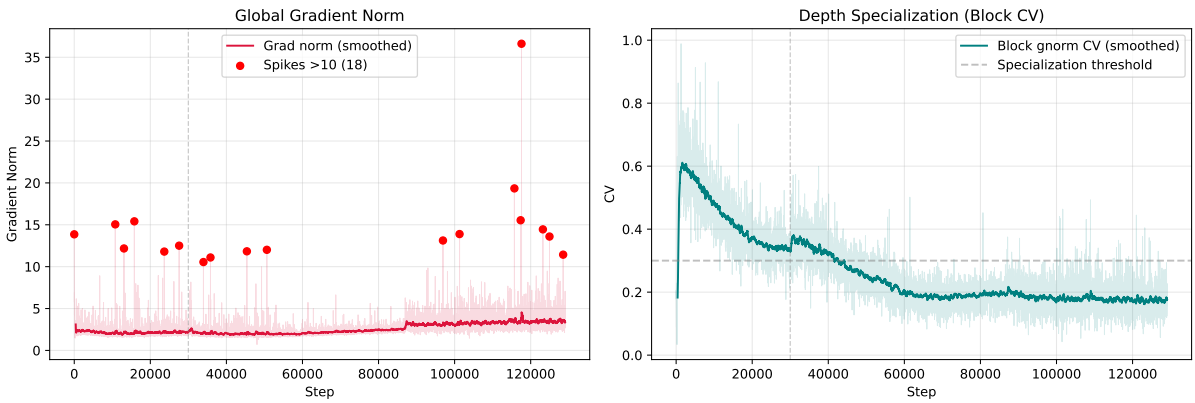


Figure 3: Global gradient norm and block CV over 130K steps. The phase boundary produces a gradient norm spike. Block CV drops from 0.55 to 0.18 — depth specialization decreases over time.

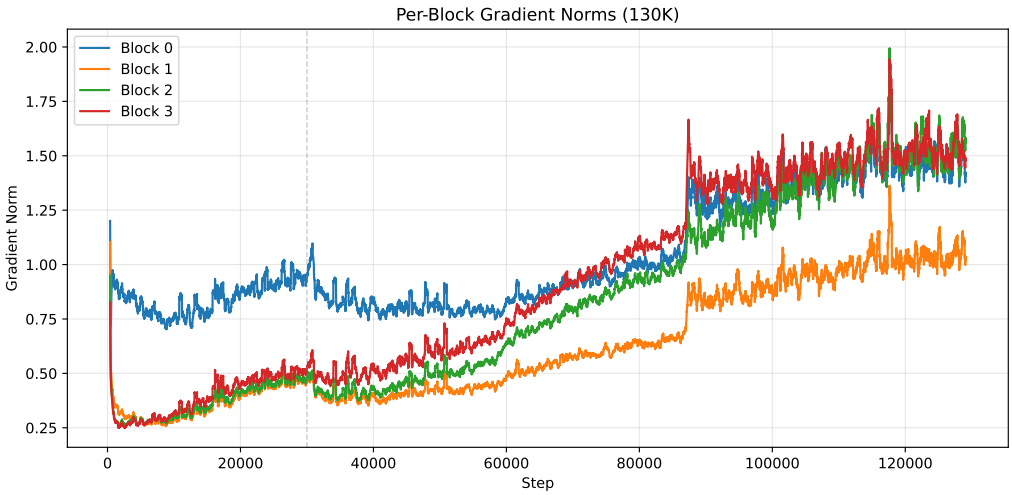


Figure 4: Per-block gradient norms. The blocks converge toward each other during phase 2, consistent with the falling CV.

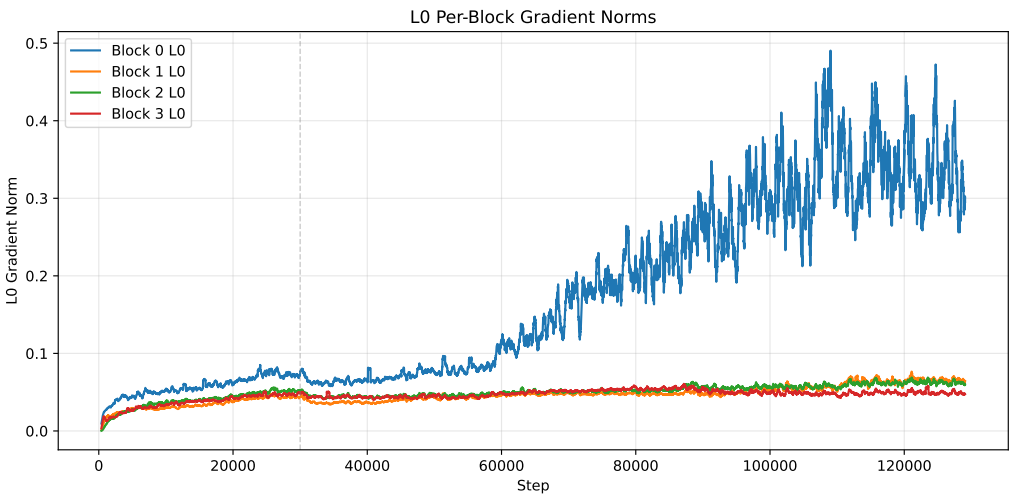


Figure 5: L0 per-block gradient norms. Block 0 continues to carry the strongest L0 gradient signal.

### 3.1 Depth Specialization Collapse

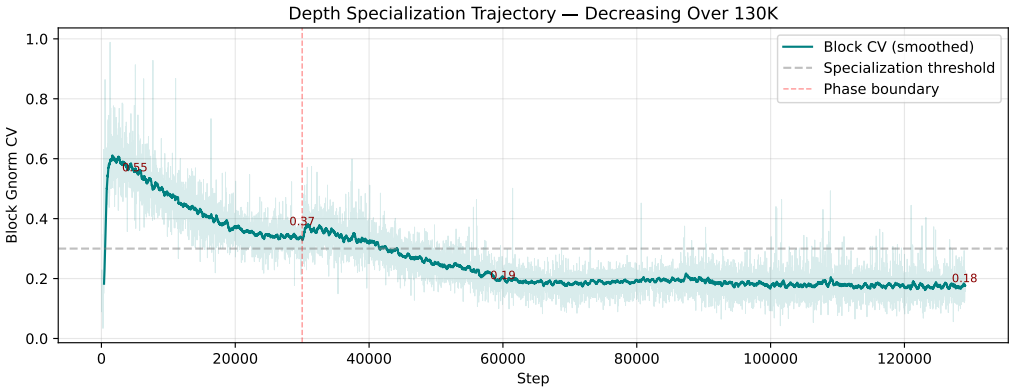


Figure 6: Block gnorm CV trajectory with annotated values. The CV drops below the 0.3 specialization threshold by step ∼45K and continues declining to 0.18 at step 129K.

Table 3: Block CV at milestones.

| Step | Block CV |
|---:|---|
| 5,000 | 0.549 |
| 30,000 | 0.368 |
| 60,000 | 0.188 |
| 100,000 | 0.181 |
| 129,000 | 0.176 |

The CV drops from 0.55 (phase 1 early, strong specialization) to 0.18 (phase 2 late, minimal specialization). This is a qualitative shift: at d=512 k=2, extended training causes blocks to **homogenize** their gradient norms. This contrasts with d=1024 (shakedown report), where CV reached 0.84–0.92. Depth specialization appears to be model-size–dependent.

### 3.2 Gradient Spikes

Table 4: All 18 gradient spikes (>10). Phase 2 has more spikes, including a 36.6-norm event.

| Step | Grad Norm | Phase |
|---|---|---|
| 0 | 13.86 | Init |
| 10,784 | 15.05 | Phase 1 |
| 13,072 | 12.17 | Phase 1 |
| 15,800 | 15.41 | Phase 1 |
| 23,672 | 11.80 | Phase 1 |
| 27,576 | 12.51 | Phase 1 |
| 33,968 | 10.55 | Phase 2 (early) |
| 35,816 | 11.10 | Phase 2 |
| 45,384 | 11.83 | Phase 2 |
| 50,640 | 12.01 | Phase 2 |
| 96,936 | 13.12 | Phase 2 |
| 101,296 | 13.89 | Phase 2 |
| 115,720 | 19.34 | Phase 2 (late) |
| 117,360 | 15.54 | Phase 2 |
| **117,584** | **36.62** | **Phase 2 (max)** |
| 123,208 | 14.45 | Phase 2 |
| 124,944 | 13.60 | Phase 2 |
| 128,536 | 11.43 | Phase 2 |

Phase 1 had 6 spikes (consistent with the 30K-only report). Phase 2 adds 12 more, with the late-training cluster (steps 115K–128K) containing the largest spike (36.6 at step 117,584). The model does not diverge after any spike — gradient clipping (max_grad_norm=1.0) contains all of them.

## 4 Gate Diagnostics: The Headline Finding

This section documents the three correlated phenomena that define the extended proxy training signature: L1 eta collapse, L1 alpha floor-pinning, and L0 eta saturation.

### 4.1 L1 Eta Collapse (The Headline)

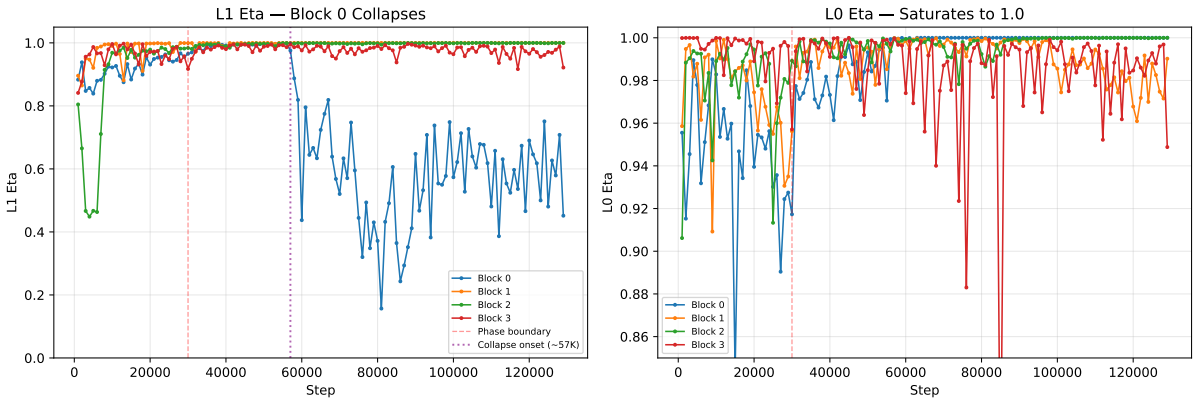


Figure 7: **The headline finding.** Left: Block 0's L1 eta drops from ∼0.96 to oscillating 0.15–0.75 after step ∼57K. Other blocks' L1 eta remains near 1.0. Right: L0 eta saturates to 1.0 across all blocks by step 57K.

Table 5: Block 0 L1 eta at key milestones.

| Step | B0 L1 Eta |
|---|---|
| 1,000 | 0.883 |
| 10,000 | 0.922 |
| 29,000 | 0.955 |
| 50,000 | 0.996 |
| 75,000 | 0.445 |
| 100,000 | 0.574 |
| 125,000 | 0.480 |

**Interpretation**: Under proxy gradients, L1's inner-loop learning rate (theta) stays near zero (Blocks 0/1/3), so L1's memory barely updates. Over 130K steps, the outer-loop optimizer learns that L1's output contribution is not worth routing through — it suppresses the output gate. Block 0 is most affected because it is the first block and receives the strongest gradient signal for gate adjustment.

This is **not** a training failure — loss continues to improve. It is an adaptive response: the model routes information through L0 (which has exact gradients and theta=1.0) and progressively shuts down L1's contribution in Block 0. Blocks 1–3 maintain high L1 eta (>0.95) because their L1 signal, while also proxy-derived, is less redundant with their L0.

## 4.2 L1 Alpha Floor-Pinning

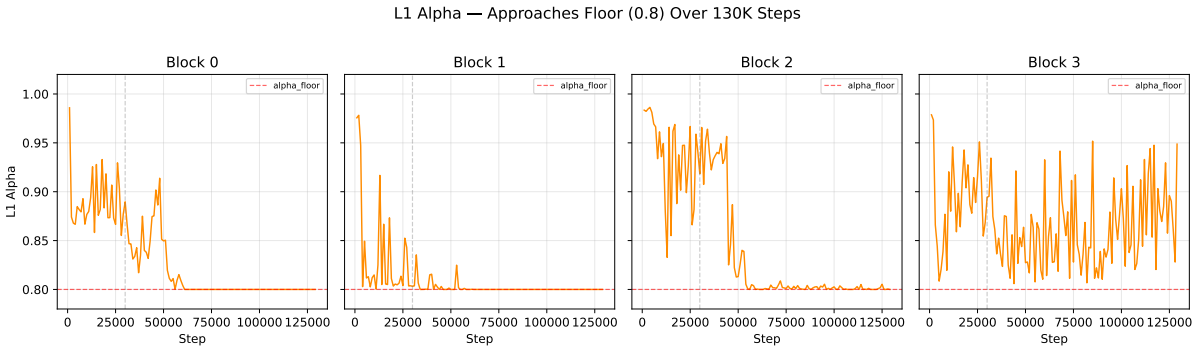


Figure 8: L1 alpha across all blocks. Blocks 0, 1, and 2 reach the alpha_floor (0.80) by step 60K. Block 3 oscillates between 0.83 and 0.96.

L1 alpha (retention) decreasing to the floor means the model wants L1 to **forget faster**. Combined with low L1 theta (slow learning), this produces an L1 that barely writes new information and rapidly forgets what it has. The floor constraint (0.80) prevents complete L1 memory erasure.

## 4.3 L0 Eta Saturation

L0 eta saturates to 1.0 in all blocks by step ∼57K (see Figure 5, right panel). This is the complementary signal to L1 eta collapse: as L1's output gate closes, L0's opens fully. The model commits entirely to L0 for output contribution, confirming L0 as the dominant level under proxy training.

## 4.4 Full Gate Evolution

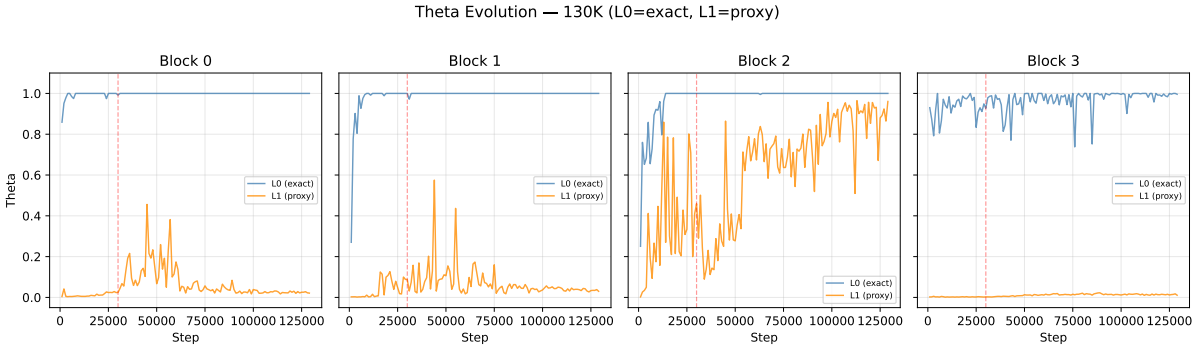


Figure 9: Theta evolution over 130K steps. L0 theta saturates to 1.0 (all blocks). L1 theta stays near zero (Blocks 0/1/3) except Block 2, which rises to 0.88.

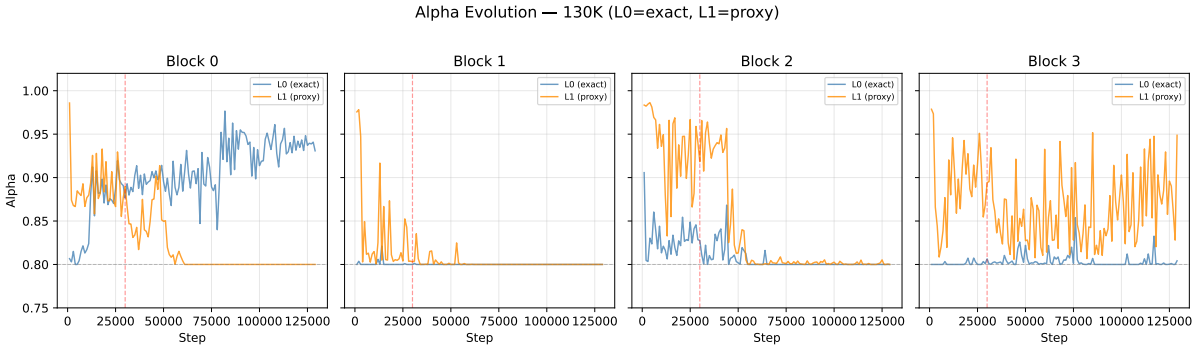


Figure 10: Alpha evolution. L0 alpha is mixed (Block 0 rises to 0.94, others near floor). L1 alpha trends toward the 0.80 floor in all blocks.

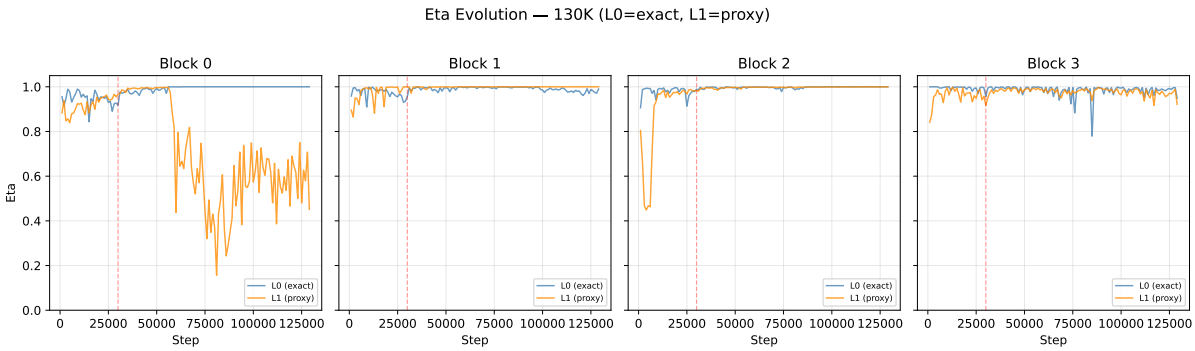


Figure 11: Eta evolution. Note the eta y-axis starts at 0.0 to show the full collapse range.

### 4.5 Block 2 L1 Theta: The Persistent Anomaly

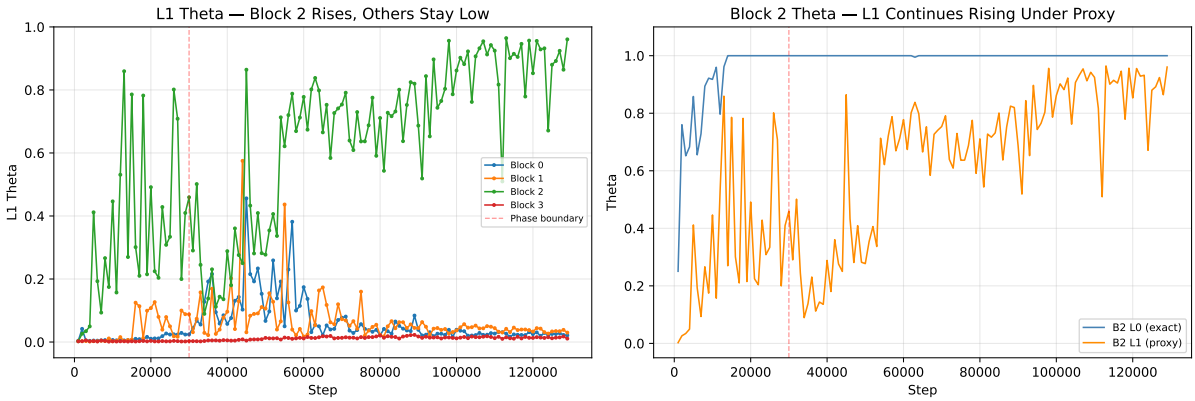


Figure 12: Left: L1 theta across blocks — Block 2 diverges dramatically from the others. Right: Block 2's L1 theta continues rising from 0.45 (30K) to 0.88 (125K) under proxy gradients, while L0 theta saturates to 1.0.

Block 2 is the only block where L1 proxy theta increases beyond 0.5. This anomaly appeared in the 30K-only spec27 report ($\theta = 0.41$), and it **intensifies** under extended training ($\theta = 0.88$). The same block was anomalous at d=1024 in the shakedown ($\theta = 0.51$). This suggests a depth-specific property of Block 2's position in the 4-block architecture: it receives sufficient gradient signal through the output path alone to drive L1 learning, even without the M trajectory chain that proxy truncates.

## 5 Detailed Gate Comparison Table

Table 6: Gate values at four milestones across the 130K run.

| | Step 1K | | | Step 29K | | | Step 75K | | | Step 125K | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **B.L** | $\alpha$ | $\theta$ | $\eta$ | $\alpha$ | $\theta$ | $\eta$ | $\alpha$ | $\theta$ | $\eta$ | $\alpha$ | $\theta$ | $\eta$ |
| B0L0 | .81 | .86 | .96 | .89 | 1.0 | .93 | .89 | 1.0 | 1.0 | .94 | 1.0 | 1.0 |
| B0L1 | .99 | .00 | .88 | .88 | .02 | .96 | .80 | .06 | .44 | .80 | .03 | .48 |
| B1L0 | .80 | .27 | .96 | .80 | 1.0 | .94 | .80 | 1.0 | 1.0 | .80 | 1.0 | .99 |
| B1L1 | .98 | .00 | .90 | .80 | .09 | 1.0 | .80 | .16 | 1.0 | .80 | .03 | 1.0 |
| B2L0 | .91 | .25 | .91 | .83 | 1.0 | .98 | .80 | 1.0 | 1.0 | .80 | 1.0 | 1.0 |
| B2L1 | .98 | .00 | .80 | .94 | .41 | .98 | .80 | .64 | 1.0 | .81 | .88 | 1.0 |
| B3L0 | .80 | .93 | 1.0 | .80 | .95 | 1.0 | .80 | 1.0 | 1.0 | .80 | 1.0 | .99 |
| B3L1 | .98 | .00 | .84 | .87 | .00 | .96 | .83 | .02 | .98 | .90 | .01 | .97 |

Reading the table column by column reveals the 130K trajectory:

- **Step 1K**: L1 alpha high (0.98–0.99), L1 theta near zero, L1 eta moderate. L1 is retaining initialization and not learning.
- **Step 29K**: L1 alpha drops to 0.80–0.94 (approaching floor). Block 2 L1 theta reaches 0.41. Other L1 theta still near zero.
- **Step 75K**: L1 alpha at floor in Blocks 0–2. Block 2 L1 theta reaches 0.64. **Block 0 L1 eta collapses to 0.44.** L0 eta saturates to 1.0 in all blocks.
- **Step 125K**: Block 2 L1 theta reaches 0.88. Block 0 L1 eta remains collapsed (0.48). The pattern is stable — no recovery.

# 6 Throughput

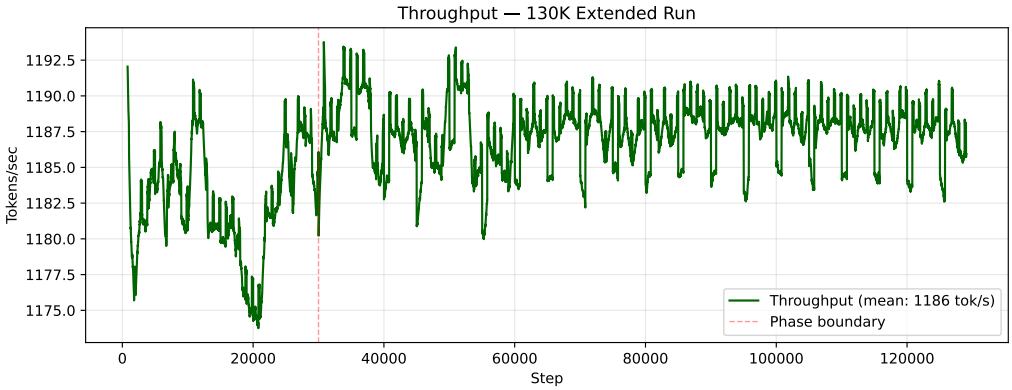


Figure 13: Throughput over 130K steps. Mean: 1,186 tok/s. Consistent across both phases with no degradation.

Total wall time: 11.88 hours (3.61h phase 1 + 8.27h phase 2). The throughput is consistent with the 30K-only spec27 report (1,182 tok/s) and spec25 (1,207 tok/s). The proxy strategy does not degrade throughput at k=2, even over 130K steps.

# 7 Conclusions

## 7.1 Extended Proxy Training Produces Adaptive L1 Suppression

The 130K run reveals that proxy gradients at L1 create a **long-timescale adaptive response** not visible at 30K steps. The model does not diverge or degrade — it adapts by:

1. Saturating L0 eta to 1.0 (commit to the exact-gradient level).
2. Collapsing Block 0 L1 eta (suppress the most redundant proxy-level contribution).
3. Pinning L1 alpha to the floor (forget faster, reducing stale proxy-informed memory).
4. Keeping L1 theta near zero (minimal proxy-gradient inner-loop updates).

This is a coherent strategy: the model learns that L1's proxy gradients are less informative than L0's exact gradients, and routes information accordingly. The loss continues to improve (2.72 at 30K $\rightarrow$ 2.60 at 129K), so this is **not degradation but optimization**.

## 7.2 Block 2: The Exception That Proves the Rule

Block 2 L1 is the only position where proxy theta rises significantly (to 0.88). This block's L1 receives sufficient gradient signal through the output path alone to sustain learning. Its L1 eta stays at 1.0 (not suppressed), and its L1 alpha reaches the floor (maximum throughput of new information). Block 2 L1 is effectively an "active proxy learner" — the rest are "passive retainers."

## 7.3 Implications for k=4 Production

At k=4, Levels 1–3 would all use proxy gradients. The 130K results predict:

- **Levels 1–3 will be suppressed over time** (low theta, declining eta), except in specific blocks (like Block 2) that receive sufficient output-path gradient.

- **L0 will dominate** — the model will route most information through the one exact-gradient level.
- **Loss will still improve** — the model adapts, it doesn't break.
- **The alpha floor is load-bearing** — without it, L1 alpha would likely drop below 0.80, causing faster forgetting and potentially worse loss. The floor constrains the suppression.

The key question for k=4 is whether the **proxy levels add value beyond what L0 alone provides**. The 130K results suggest they do (loss 2.60 vs single-level k=1 baselines at ∼3.0), but the value may be primarily from the retained memory (high alpha) rather than active learning (low theta). This is a "memory buffer" role rather than the "active learner" role that exact gradients produce.

### 7.4 Depth Specialization at d=512

The Block CV collapse from 0.55 to 0.18 indicates that extended training at d=512 leads to block homogenization. This is the opposite of d=1024 (shakedown: CV 0.84–0.92), suggesting that:

- Depth specialization is model-size–dependent.
- At d=512, 4 blocks may be more than needed — the model converges to a uniform representation.
- At d=1024+, blocks have enough capacity to specialize into distinct roles (stateless processor, gated specialist, full engagement).

### 7.5 Next Steps

1. **Shard M diff logging**: Implement the Spec 27 sentinel metric ($\|M_{\text{final}} - M_{\text{start}}\|_F$) to quantify proxy-level learning without the DGD diagnostic.
2. **Ablation: alpha_floor removal at 130K**: Test whether removing the floor allows complete L1 suppression and how this affects loss.
3. **k=4 d=512 extended**: Run the push-up pipeline to k=4 with proxy at L1–L3, observing whether multiple proxy levels suppress in sequence.
4. **d=1024 k=2 extended**: Replicate this 130K experiment at d=1024 to test whether L1 eta collapse also occurs when depth specialization is strong.